

Pressure Matching With Forced Filters for Personal Sound Zones Application

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This paper presents a rethinking of the Pressure Matching Method (PM) used in the generation of Personal Sound Zones when the responses of some filters are already known. They are then imposed in the calculation, resulting in a Forced Pressure Matching method. This new formulation is implemented to control two zones—a reproduction zone and dark zone—in a two-seat configuration aimed toward the Transportation industry. Due to variations in transportation acoustic environments, the computational time is added to the metrics typically used in the Personal Sound Zones literature (such as acoustic contrast, effort, error, etc.), foreseeing the need of an adaptive system. Perfect Dirac delta functions were forced as filters of the loudspeakers closest to the reproduction zone. The new formulation achieved the same acoustic contrast, effort, and reproduction error very similar to that of the conventional PM but calculated the filters 24% faster.

0 INTRODUCTION

The generation of Personal Sound Zones (PSZ) has been a topic of great interest for more than twenty years [1]. A PSZ system's minimum requirement is to be able to reproduce a specific program over a so-called bright zone (BZ) while minimizing it in a different area, a denoted dark zone (DZ) without the use of headphones or any insulating panels. Moreover, it is possible to assume that by placing a finite number of acoustic sources (L), e.g., loudspeakers, the pressure field in each zone will be the sum of the radiated pressure of every source. Hence, if a filter unit is placed at the input of each source, it is possible to calculate the optimal combination of filters to control the desired zones.

The least-squares method, also known as pressure matching (LSM/PM), has been widely used in the generation of PSZ since Poletti [2] in 2008 (previously used in other applications like Sound Field Synthesis) because it offers an explicit expression of the filters that minimize the least-squared error (LSE) between the reproduced field and a "desired" pressure field, better known as the target pressure (p_T). Moreover, in 2002, Choi et al. stated that the magni-

tude of a sound perception is determined by its potential energy [3]. This was later adopted in the PM methods dividing the PSZ into a twofold problem throughout the definition of the p_T : an LSE minimization in the BZ and a potential energy minimization in the DZ [4–6].

To this day, the PSZ algorithms consider every filter as a possible variable to calculate which obliges the system to have the capabilities either to calculate or store all the solutions. However, it might not be necessary to consider every single filter as a variable to calculate. For example, it could be seen as part of the design constraint; i.e., to impose a specific response into some of the available filters either due to limitations in the transducers or to ensure the quality of the filter, etc. Furthermore, in some cases it is possible to know in advance the resulting shape of some filters [7], i.e., a piece of the solution is already known and there is no need to calculate it again.

One of the main fields of application of the PSZ is the transportation industry (TI) [8–10], which presents not only complex acoustic environments (small cavities, a mix of very reflective and very absorbent surfaces, etc.) but also dynamic acoustic environments in constant change (e.g.,

changes in temperature [11], window opening, number of passengers, seat positioning, etc.). Hence the presence of an adaptive system might be necessary, implying that the computational cost has to be taken into account as a design constraint.

In 2019, Vindrola et al. [7] studied a custom-made headrest in a two-seat configuration aiming toward the TI. It was observed that the filters associated to the sources closer to the BZ were almost perfect unit-sample/Dirac delta functions (8). This result was obtained because their contribution to the reproduction is mainly direct field—much closer to the BZ—and the fact that the p_T is defined to be these two sources driven in phase (all these aspects are re-addressed in this paper).

Therefore, not only these particular sources can be driven directly without any filtering, but there is no need to calculate its associated filters to begin with, thus reducing the number of unknown variables for the system to calculate with a minimum performance compromise. However, their contribution to the overall problem cannot simply be removed or ignored. Moreover, in 2006 Elliott et al. [8] studied a TI system using two loudspeakers installed in one side of a headrest. In their work, both filters were calculated and normalized by the frequency response of the one in the reproduction side (i.e., the BZ), allowing them to directly drive the source in the reproduction side and to have one “cancellation filter.” In spite of the fact that this approach reduces the number of filters to be adapted—as shown in the paper only one adaptive filter is needed—it does not take into account the effect of the cancellation system in the quality of the reproduced field in the BZ.

The dimensional aspects of the system (loudspeaker and microphones arrays) and the wavelength of the sound determine the frequency boundaries where a technology is capable of generating PSZ [1]. Therefore, in this study, the methods are proposed to control the frequency band between 50–1,000 Hz, and it is assumed that the higher frequencies are controlled using a different technology (e.g., beamforming [12, 13]). Nevertheless, it is important to highlight that the theoretical procedure introduced in this paper is not limited to this bandwidth.

The aim of this paper is to introduce the Forced Pressure Matching (FPM), a rethinking of the classical LSM/PM algorithm where the responses of one or more filters are forced in the calculation. Aiming toward a TI application, the FPM is compared with the PM algorithm using two adjacent seats, each one equipped with four loudspeakers in the headrest. To quantify their performance, five metrics are used (Sec. 2.1): the Acoustic Contrast (AC), Effort (E_{ff}), condition number, normalized Reproduction Error (E_{rr}), and calculation time. Lastly, the robustness of the algorithms is tested by adding white noise to the impulse responses.

The paper is constructed as follows: firstly in Sec. 1, a review on the PM is done, explaining the formulation of the FPM. In Sec. 2, the system under study is presented as well as the parameters used in the calculation. In Sec. 3, the analytical implications of the Target pressure definition in both formulations is reviewed. Then, in Sec. 4, the

algorithms are implemented using real impulse responses (IR) measured under anechoic conditions. Lastly in Sec. 5, the main conclusions are summarized and further works are proposed.

1 THEORY

In this section, the description of the acoustical problem is introduced, the formulations of PM and FPM are explained, and the metrics used for the evaluation of the performance are defined.

The aim of the PSZ methods is to obtain the best filter combination that solves two problems simultaneously: one in the BZ and one in the DZ. For this paper, the methods based on constrained optimization (CO) are analyzed due to their popularity in generating PSZ, but it is necessary to highlight that there are other methods—for example the methods based on spatial filtering [14].

The CO methods used for PSZ are based in defining a cost function ($\mathcal{L}$) that tackles the problem in the BZ and DZ. Several expressions of $\mathcal{L}$ have been proposed for the generation of PSZ: Acoustic Contrast Control [3], Acoustic Planarity Control [15], Broadband Acoustic Contrast Control with a Response Trend Estimation [16], Acoustic Contrast with Limited Sound Differences [17], etc. For this paper, the PM formulation is chosen.

In this study, the frequency domain formulation of the PSZ problem is used not only because it is the most widely used to tackle PSZ problems (e.g., [9, 3, 18] among many others) but also because it is the one that presents the smallest calculation times of all the available formulations [7]—at least for filters whose length is smaller than 1,000 taps [19]. Nevertheless, it is necessary to mention that there are at least three other ways (to the best of our knowledge) for which the same principles introduced here should be applicable: in the time domain as convolution matrices [5, 20], using rational matrices [21], or as an optimal filtering problem [22].

1.1 Classical Pressure Matching Formulation

Assuming that the BZ and DZ are both sampled with K microphones each, the pressure p at the microphone k in the BZ can be expressed as in Eq. (1) for a single angular frequency (ω)

$$p_{Bk}(\omega) = \sum_{l=1}^L G_{Bkl}(\omega) q_l(\omega), \quad (1)$$

being q_l the complex weight (CW) applied to the source l at the frequency ω and $G_{Bkl}(\omega)$ the transfer function between the microphone k and the source l . The same development can be used analogously to describe the DZ. For simplicity the ω term is ignored.

From Eq. (1) it is possible to express the pressure for all the K microphone positions using the matrix form as

$$\mathbf{p}_B = \mathbf{G}_B \mathbf{q} \quad \text{and} \quad \mathbf{p}_D = \mathbf{G}_D \mathbf{q}, \quad (2)$$

where $\mathbf{p}_B$ and $\mathbf{p}_D$ are the $K \times 1$ vectors of complex pressures in the BZ and DZ, respectively, $\mathbf{G}_B$ and $\mathbf{G}_D$ are the $K \times L$ matrices of complex transfer functions (TF), and

$$\mathbf{q} = [q_1, q_2, \dots, q_L]^T \quad (3)$$

is the $(L \times 1)$ vector of complex weights, T being the transpose operator.

Two vectors ($\mathbf{e}_B$ and $\mathbf{e}_D$) can be used to express the error on every microphone in the BZ and DZ, respectively, as the difference between the vector of target pressure field ($\mathbf{p}_T$) and the sum of the contribution of each filtered source as

$$\mathbf{e} = \begin{bmatrix} \mathbf{e}_B \\ \mathbf{e}_D \end{bmatrix} = \begin{bmatrix} \mathbf{p}_{TB} \\ \mathbf{p}_{TD} \end{bmatrix} - \begin{bmatrix} \mathbf{G}_B \\ \mathbf{G}_D \end{bmatrix} \mathbf{q} \quad (4)$$

where $\mathbf{p}_{TB}$ and $\mathbf{p}_{TD}$ are the $K \times 1$ target pressure vectors in the BZ and DZ, respectively.

Furthermore, the cost function ($\mathcal{L}$) in the pressure matching methods is defined as the LSE containing both the BZ_{Term} and DZ_{Term}

$$\begin{aligned} \mathcal{L} &= |\mathbf{e}|^2 = |\mathbf{e}_B|^2 + |\mathbf{e}_D|^2 \\ &= (\mathbf{p}_{TB} - \mathbf{G}_B \mathbf{q})^H (\mathbf{p}_{TB} - \mathbf{G}_B \mathbf{q}) \\ &\quad + (\mathbf{p}_{TD} - \mathbf{G}_D \mathbf{q})^H (\mathbf{p}_{TD} - \mathbf{G}_D \mathbf{q}) \end{aligned} \quad (5)$$

H being the Hermitian operator.

Lastly it is possible to obtain the expression of the complex weights—following the Lagrange multipliers method—by taking the gradient of the cost function and equating it to zero ($\nabla \mathcal{L} = \mathbf{0}$) as:

$$\mathbf{q} = (\mathbf{G}_B^H \mathbf{G}_B + \mathbf{G}_D^H \mathbf{G}_D)^{-1} (\mathbf{G}_B^H \mathbf{p}_{TB} + \mathbf{G}_D^H \mathbf{p}_{TD}), \quad (6)$$

being $\mathbf{0}$ a vector full of zeros. Eq. (6) is the classical expression of the complex weights that can be found in the literature and it is valid as long as $2K > L$ [2].

1.2 Forced-Filters Pressure Matching

In the classical PM formulation, every filter is considered as a variable that needs to be calculated. However, there are some cases where it might not be necessary to recalculate every filter; e.g., a specific response wants to be applied to one source (or more) or if the response of some filters is known in advance.

Hence, assuming that among $\mathbf{q}$ there are N weights ($\mathbf{q}_f$) one wants to force and $L - N$ weights ($\mathbf{q}_c$) to calculate, the vector of CW can be expressed as:

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_c \\ \mathbf{q}_f \end{bmatrix} = \{[q_{c1}, \dots, q_{c(L-N)}][q_{f1}, \dots, q_{fN}]\}^T \quad (7)$$

where $0 < N < L$; otherwise if $N = 0$ is the classical PM and if $N = L$ all the filters are known.

Furthermore, in $\mathbf{G}_B$ and $\mathbf{G}_D$ there are some TF that correspond to the loudspeakers associated to the forced filters and others that correspond to the filters to calculate:

$$\begin{bmatrix} \mathbf{G}_B \\ \mathbf{G}_D \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{Bc} & \mathbf{G}_{Bf} \\ \mathbf{G}_{Dc} & \mathbf{G}_{Df} \end{bmatrix}, \quad (8)$$

where $\mathbf{G}_{Bc}$ and $\mathbf{G}_{Dc}$ are the $K \times (L - N)$ matrices of complex TF that correspond to the CW to calculate in the

BZ and DZ, respectively. $\mathbf{G}_{Bf}$ and $\mathbf{G}_{Df}$ are the $K \times N$ matrices of complex TF that correspond to the forced CW in the BZ and DZ, respectively.

Introducing Eq. (7) and Eq. (8) in the reproduction error (Eq. (4)) one obtains

$$\mathbf{e} = \begin{bmatrix} \mathbf{p}_{TB} \\ \mathbf{p}_{TD} \end{bmatrix} - \begin{bmatrix} \mathbf{G}_{Bc} & \mathbf{G}_{Bf} \\ \mathbf{G}_{Dc} & \mathbf{G}_{Df} \end{bmatrix} \begin{bmatrix} \mathbf{q}_c \\ \mathbf{q}_f \end{bmatrix}. \quad (9)$$

Furthermore, the resulting pressure of the forced CW can be calculated as follows

$$\begin{bmatrix} \mathbf{p}_{Bf} \\ \mathbf{p}_{Df} \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{Bf} \\ \mathbf{G}_{Df} \end{bmatrix} \mathbf{q}_f \quad (10)$$

and subtracted from the original $\mathbf{p}_T$ one obtains:

$$\mathbf{e}_c = \begin{bmatrix} \mathbf{p}_{TB} - \mathbf{p}_{Bf} \\ \mathbf{p}_{TD} - \mathbf{p}_{Df} \end{bmatrix} - \begin{bmatrix} \mathbf{G}_{Bc} \\ \mathbf{G}_{Dc} \end{bmatrix} \mathbf{q}_c. \quad (11)$$

From Eq. (11) a new target pressure can be defined ($\mathbf{p}_{Tc}$) containing both the information from the original $\mathbf{p}_T$ and the information from the forced filters.

$$\mathbf{e}_c = \begin{bmatrix} \mathbf{p}_{TBc} \\ \mathbf{p}_{TDc} \end{bmatrix} - \begin{bmatrix} \mathbf{G}_{Bc} \\ \mathbf{G}_{Dc} \end{bmatrix} \mathbf{q}_c \quad (12)$$

As in Sec. 1.1, with the new error introduced in Eq. (12), both the new cost function ($\mathcal{L}_c$)

$$\begin{aligned} \mathcal{L}_c &= |\mathbf{e}_c|^2 = |\mathbf{e}_{Bc}|^2 + |\mathbf{e}_{Dc}|^2 \\ &= (\mathbf{p}_{TBc} - \mathbf{G}_{Bc} \mathbf{q}_c)^H (\mathbf{p}_{TBc} - \mathbf{G}_{Bc} \mathbf{q}_c) \\ &\quad + (\mathbf{p}_{TDc} - \mathbf{G}_{Dc} \mathbf{q}_c)^H (\mathbf{p}_{TDc} - \mathbf{G}_{Dc} \mathbf{q}_c) \end{aligned} \quad (13)$$

and the expression of $\mathbf{q}_c$:

$$\mathbf{q}_c = (\mathbf{G}_{Bc}^H \mathbf{G}_{Bc} + \mathbf{G}_{Dc}^H \mathbf{G}_{Dc})^{-1} (\mathbf{G}_{Bc}^H \mathbf{p}_{TBc} + \mathbf{G}_{Dc}^H \mathbf{p}_{TDc}) \quad (14)$$

can be calculated.

It is important to highlight, despite its redundancy, that there is no need to re-calculate the forced filters ($\mathbf{q}_f$), so the number of CW to calculate (i.e., the filters) as well as the size of the complex transfer functions matrices involved in the calculation are reduced.

2 SYSTEM UNDER STUDY

The system under study is a typical transportation industry application. Two listening positions are separated by 73 cm. Foreseeing the presence of a listener, the zones are discretized with 24 measuring positions per seat uniformly distributed in the plane of the ears. These are divided into 2 arrays of 12 points per ear (see Fig. 1).

Nowadays, many manufacturers in the TI use in their products two loudspeakers in the headrest—especially in cars. On the one hand, the nearfield arrays allow them to extend the bandwidth of the system and improve the coupling between the microphones and the control sources and the response is mainly dominated by the direct sound [23]. On the other hand, the main drawback is that the effects of objects in close proximity to the sources may have a significant effect on the performance [23].

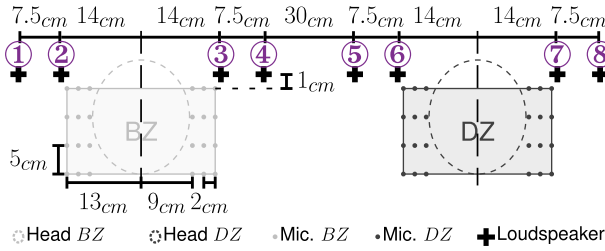


Fig. 1. Symmetric geometry composed by 8 Sources and 48 Microphone positions.

In order to enhance the AC achieved by the system, one pair of sources (4 and 5) is added between the seats resembling the studies made by Elliott et al. in 2006 [8] or Yanagidate et al. in 2014 [17]. In addition, a second pair is installed (1 and 8) in the outer sides of the system, foreseeing the presence of reflective panels or the possibility to add more listening zones. The system under study aims toward a Transportation industry application; however the FPM algorithm could be applied in other architectures.

The Impulse Responses are sampled at a sampling frequency of $F_s = 2,450$ Hz and the length of the IR used for the calculations is set to 0.16 s (giving a frequency step of 6.25 Hz for the complete bandwidth). The frequency vector used for the calculation is the one obtained from the aforementioned IR; i.e., from 0 Hz to the Nyquist frequency (1,225 Hz) with a resolution of 6.25 Hz. Regarding the output filter, the same frequency resolution is used and an inverse Fourier transform is applied to the resulting complex weights without any extra windowing, interpolation, filtering, etc. All the calculations were performed using MATLAB software, under Windows 7 Enterprise, using an Intel Core(TM) i7-6820HQ CPU with a RAM memory of 16 GB and a 64-bit architecture.

2.1 Metrics

In order to compare the performance of the formulations, five metrics are introduced and defined hereafter.

2.1.1 Acoustic Contrast

In 2002, Choi et al. [3] stated that the acoustic potential energy determines the magnitude of sound perception. Therefore to measure the “acoustic separation” between the zones it is possible to compare the acoustic potential energy between the two desired zones, i.e., the Acoustic Contrast (AC). Hence, assuming both zones have the same dimensions, the AC can be expressed for a single frequency (ω) as

$$AC = \frac{\mathbf{q}^H \mathbf{G}_B^H \mathbf{G}_B \mathbf{q}}{\mathbf{q}^H \mathbf{G}_D^H \mathbf{G}_D \mathbf{q}}. \quad (15)$$

The AC is the most widely used metric to evaluate the “separation” between both zones and it is generally expressed in dB.

2.1.2 Condition Number

In computing problems, such as optimization, ill-conditioning may describe any computation whose output values are very sensitive to small changes in the input data [10]. In the case of having to perform matrix inversions (as in Eqs. (6) and (14)), it is necessary to be certain that the matrix is not ill-conditioned. The normally used condition number of a symmetric matrix with respect to its inverse—as will be the case in this paper—is given by the ratio of its largest to its smallest eigenvalue [18]. A large condition number indicates that the matrix is close to singular, i.e., the matrix is ill-conditioned [10].

2.1.3 Effort

The E_{ff} is defined for a single frequency (ω) as

$$E_{ff} = 10 \log \left(\frac{\mathbf{q}^H \mathbf{q}}{|\mathbf{q}_r|^2} \right), \quad (16)$$

where $\mathbf{q}$ is the calculated complex weights and $\mathbf{q}_r$ is the weight necessary to produce the same Sound Pressure Level in the BZ [18] when driving sources 2 and 3 in phase, becoming an indicator of the system’s power consumption [24].

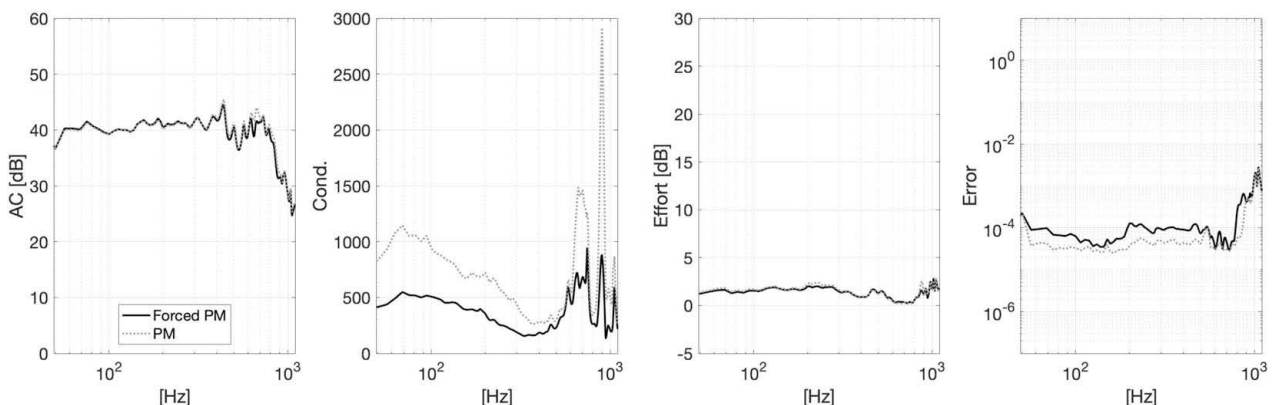


Fig. 2. Resulting AC, Cond., E_{ff} , and E_{rr} for the FPM (bold) and the PM (dashed) using measured transfer functions.

2.1.4 Normalized Reproduction Error

Another aspect that is necessary to evaluate is the quality of the reproduced field. As the aim of the system is to reproduce the $\mathbf{p}_{TB}$, it is possible to measure how far from this target field the system is by measuring the normalized error E_{rr} [9]. Therefore the error can be expressed as

$$E_{rr} = \frac{(\mathbf{p}_{TB} - \mathbf{p}_B)^H (\mathbf{p}_{TB} - \mathbf{p}_B)}{\mathbf{p}_{TB}^H \mathbf{p}_{TB}} \quad (17)$$

2.1.5 Calculation Time

It is defined in this paper—assuming a broadband system—as the time taken by each algorithm to obtain the filter's IR from the matrix of the system's IRs. Hence, in the case of the formulation in the frequency domain, both the Fourier transform (FT)—to obtain $\mathbf{G}_B$ and $\mathbf{G}_D$ —and the Inverse Fourier Transform (IFT) are considered. The calculation time in this paper is expressed as the mean value achieved for 2×10^4 iterations of each algorithm (giving an accuracy of ± 1 millisecond, approximately).

3 THE TARGET PRESSURES

The definition of the target pressure and how it simplifies the cost function (and the resulting filters) has a direct impact on the acoustical problem trying to be solved by the algorithm. In the following, the governing equations of classical pressure matching and forced pressure matching are analyzed.

3.1 Classical Pressure Matching

As it is usually defined in the literature [5, 7, 9, 23], the $\mathbf{p}_T$ is defined in the BZ as the pressure delivered by the sources closer to the zone (② and ③) when driven in phase, or equivalently, the sum of their contribution when filtered by a δ (multiplied by 1 in the frequency domain). The $\mathbf{p}_T$ in the DZ zone is defined as a vector of zeros (see Eq. (18)). Therefore, the $\mathbf{p}_T$ for both zones can be expressed as

$$\mathbf{p}_T = \begin{cases} \mathbf{p}_{TB} = \sum_{l=2}^3 \mathbf{G}_{Bkl} \cdot 1 \\ \mathbf{p}_{TD} = \mathbf{0} \end{cases} \quad (18)$$

Implementing Eq. (18) in the cost function yields:

$$\mathcal{L} = |\mathbf{p}_{TB} - \mathbf{G}_B \mathbf{q}|^2 + |\mathbf{G}_D \mathbf{q}|^2, \quad (19)$$

with an LSE term in the BZ and an energy minimization term in the DZ—also known as one particular case of the “Combined Solution” in [4, 5]. Hence, the vector of CW becomes

$$\mathbf{q} = (\mathbf{G}_B^H \mathbf{G}_B + \mathbf{G}_D^H \mathbf{G}_D)^{-1} \mathbf{G}_B^H \mathbf{p}_{TB}. \quad (20)$$

As can be seen from Eq. (20), only the $\mathbf{p}_{TB}$ has to be defined.

3.2 Forced Pressure Matching

On the other hand, in Vindrola et al. 2019 [7] it was observed that the filters associated to the sources ② and ③ (see Fig. 1) resulted in almost perfect δ ; this observation is verified later in Fig. 4. Consequently, these are chosen

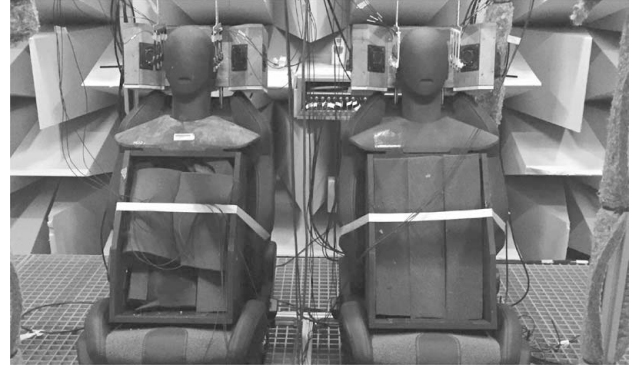


Fig. 3. Measurements at LAUM's Anechoic Chamber.

forced using the new formulation as two ($N = 2$) perfect δ —as for this paper a frequency domain formulation is used, multiplied by 1. The resulting pressure of the forced CW can be expressed as:

$$\mathbf{p}_f = \begin{cases} \mathbf{p}_{Bf} = \sum_{l=2}^3 \mathbf{G}_{Bkl} \cdot 1 \\ \mathbf{p}_{Df} = \sum_{l=2}^3 \mathbf{G}_{Dkl} \cdot 1 \end{cases} \quad (21)$$

Therefore, from Eqs. (11), (18), and (21), $\mathbf{p}_{Tc}$ can be calculated using the following equation

$$\mathbf{p}_{Tc} = [\mathbf{p}_T - \mathbf{p}_f] = \begin{cases} \mathbf{p}_{TBc} = \mathbf{0} \\ \mathbf{p}_{TDc} = -\sum_{l=2}^3 \mathbf{G}_{Dkl} \cdot 1 \end{cases} \quad (22)$$

When implementing $\mathbf{p}_{Tc}$ (Eq. (22)) in the cost function it yields:

$$\mathcal{L}_c = |\mathbf{G}_{Bc} \mathbf{q}_c|^2 + |\mathbf{p}_{TDc} - \mathbf{G}_{Dc} \mathbf{q}_c|^2 \quad (23)$$

which presents exactly the opposite behavior of that in the PM and an energy minimization problem in the BZ and an LSE problem in the DZ. Then the expression of the filters of the forced PM is given by

$$\mathbf{q}_c = (\mathbf{G}_{Bc}^H \mathbf{G}_{Bc} + \mathbf{G}_{Dc}^H \mathbf{G}_{Dc})^{-1} \mathbf{G}_{Dc}^H \mathbf{p}_{TDc} \quad (24)$$

where only a pressure in the DZ zone needs to be specified.

4 EXPERIMENTAL IMPLEMENTATION

The experimental implementation (Fig. 3) was conducted in the anechoic chamber of the LAUM using two car seats with “custom-made” plywood headrest. Each headrest was equipped with four independent closed-box loudspeakers using the Tectonic TEBM35C10-4 and an enclosure of $\simeq 1$ liter each. To introduce diffraction due to listeners, two dummy head + torso were placed on the seats. In addition, 16 Brüel & Kjaer 4958 microphones were arranged in groups of 4 per ear and moved in order to obtain the same positioning as in Fig. 1. Moreover, all the IRs were measured using a 17th-order MLS algorithm [25] averaged 24 times using an $F_s = 44,100$ Hz and 2 sets of IRs were used, one for calculation of the filters and one for the evaluation of the metrics. For the calculation of the filters, the IRs were downsampled by a factor of 18. Lastly the use of a

high-pass filter to attenuate any possible noise in the lower frequencies was ignored in this study.

4.1 Results

The optimal filter combinations were calculated with the *IRs* measured in the anechoic chamber. Therefore, to evaluate the performance of both methods one proceeded to analyze the resulting metrics, filters and their sensitivity to noise.

4.1.1 Metrics

The final performance for both methods are plotted in Fig. 2. Regarding the *AC* and E_{ff} both formulations present almost identical performances regardless of the algorithm used. On the other hand, the FPM improves the conditioning of the problem (in some frequencies by a factor of two or more). Nevertheless both formulations seem to be far from ill-conditioned, i.e., it does not seem to be necessary to add some sort of regularization term. Moreover there is a small change in performance when it comes to the E_{rr} ; the PM algorithm results in a slightly smaller reproduction error. Lastly when it comes to the computational cost, the PM took 5.5 milliseconds while the FPM took 4.2 milliseconds to calculate the filters, i.e., $\approx 24\%$ faster.

4.1.2 Filters

In order to analyze the calculated filters, the ones associated to three different loudspeakers are studied in Fig. 4.

Firstly, by observing one of the filters close to the BZ (Spk. ③ in Fig. 4), it is possible to see that both responses—the FPM and PM—are very close; the variations are less than 0.65 dB for the amplitude and less than 0.059 radians for the phase. In addition, these small discrepancies can be seen with some difficulties in the IR as a small ringing. On the other hand, when analyzing one of the filters associated to the sources closer to the DZ (Spk. ⑥ in Fig. 4), it is possible to observe that both methods result in strongly attenuated responses but also similar. The amplitude response varies less than 2.2 dB and the phase less than 0.36 radians except at 956 Hz.

However, when analyzing the response of the filter corresponding to speaker ④, the differences between the methods are larger; the amplitude response varies for some frequencies up to 6 dB and the phase up to 1 radian. Lastly the differences in the responses of the filters associated to speaker ① are considerably larger; the PM presents some notch-type depths and phase-jumps (e.g., at 280 or 656 Hz) not present in the FPM method.

By observing the speakers close to the zones ③ and ⑥, one might assume that both methods result in the same q . However, the small differences present in these sources result in bigger differences in the further sources; hence both methods result in different solutions. The repeatability of these results was evaluated using different measurements of the same system and both methods always give the same filters.

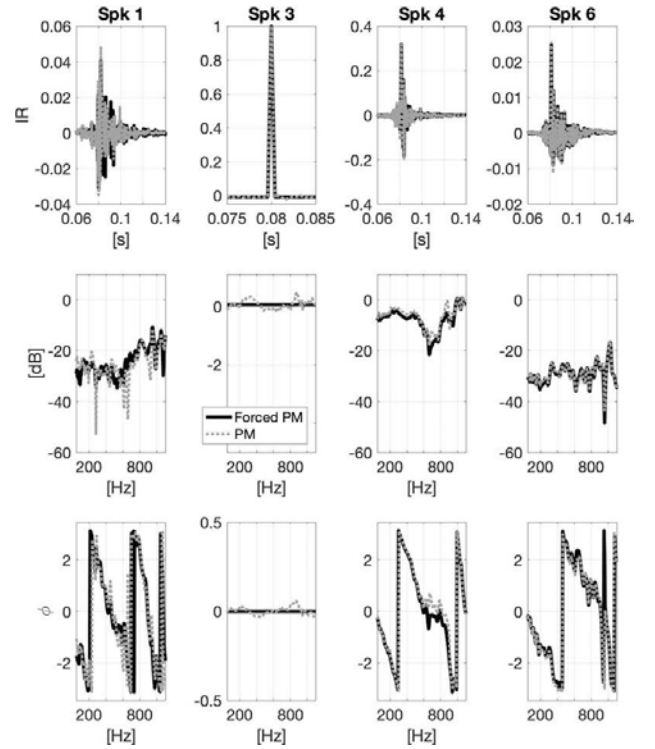


Fig. 4. Four resulting filters of the FPM (bold) and the PM (dashed) formulations using measured transfer functions. The first line corresponds to the time IR shifted to the center of the IR to be able to observe the presence of pre-ringing, the second line corresponds to the modulus of the frequency response, and the third line corresponds to the phase response.

Table 1. Table of SNR.

Case No.	Mean SNR
0	No Noise
1	35 dB
2	20 dB
3	5 dB
4	−10 dB

4.1.3 Robustness

In order to evaluate the robustness of the methods, a “noisy environment” was emulated by adding white Gaussian noise to the TFs—the ones used in the filter calculation (i.e., to solve Eqs. (20) and (24)) as well as the ones used to calculate the p_T (Eqs. (18) and (22)). Furthermore, in spite of adding a different noise to each loudspeaker-microphone pair, the same matrix of noises is added to both methods. Four different levels of noise were used, resulting in the four cases of signal-to-noise ratio (SNR) established in Table 1.

The SNR for one microphone loudspeaker combination (Bkl) is defined as

$$SNR_{Bkl} = \frac{\sum_0^F |p_{Bkl}|^2}{\sum_0^F |Noise_{Bkl}|^2}, \quad (25)$$

where $Noise_{Bkl}$ is the Noise introduced to the Bkl TF and F is the length of the TF, and analogously for the combination in the DZ.

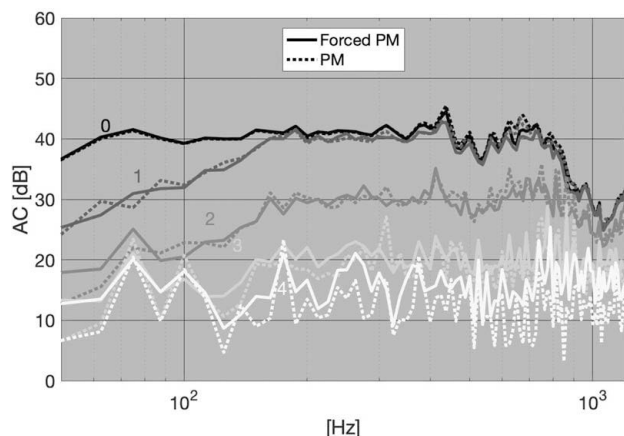


Fig. 5. Resulting AC for the FPM (bold) and the PM (dashed) using measured transfer functions for five different levels of Gaussian noise introduced in the TF as defined in Table 1.

It is necessary to highlight that the mean SNR is taken only as an indicator; the differences between the amplitudes of the TF is very large—some microphones are much closer to the loudspeaker than others. For example, in Case No. 1 the maximum SNR = 50 dB while the minimum SNR = 10 dB.

Firstly the AC was calculated with the five cases of SNR (see Fig. 5). It is possible to observe how the AC is deteriorated as the SNR is reduced. The AC performance starts deteriorating for Case No. 1 in the lowest part of the spectrum (below ≈ 120 Hz); this is due to the fact that the loudspeakers chosen for this application produce very little sound pressures at these frequencies. The other noise levels get larger to the point that in Case No. 4 there is hardly any control. Nevertheless, both formulations result in a similar deterioration for the different levels of SNR (Fig. 5).

On the other hand, when analyzing the E_{rr} , the performance changes considerably (Fig. 6). As already mentioned, without extra noise, the PM formulation results in slightly lower reproduction error. However, as soon as the SNR is reduced (in the lowest part of the spectrum first Case No. 1) this condition “flips”; the FPM outperforms the PM. Moreover when more noise is added (Case No. 2, 3, etc.), the FPM E_{rr} is always below the PM. Lastly in Case No. 4, the PM E_{rr} gets pushed close to 1 while the FPM maintains it to the order of $\approx 10^{-2}$.

This increase in the robustness is mainly due to the fact that by forcing two sources with an already known part of the solution, the algorithm tries to solve a different cost function (Eq. (23)) where the priority is in the potential energy minimization in the BZ. Moreover, when analyzing the effects of noise in the filter associated to Spk. ③ it is possible to observe (in Fig. 7) that the PM’s IR deteriorates from a δ function. This results in slightly noisier amplitudes and phase responses especially in Case No. 2. One could assume that these differences are negligible; however these small effects on the sources close to the BZ are mostly indicating the deterioration of the E_{rr} in Fig. 6. In addition,

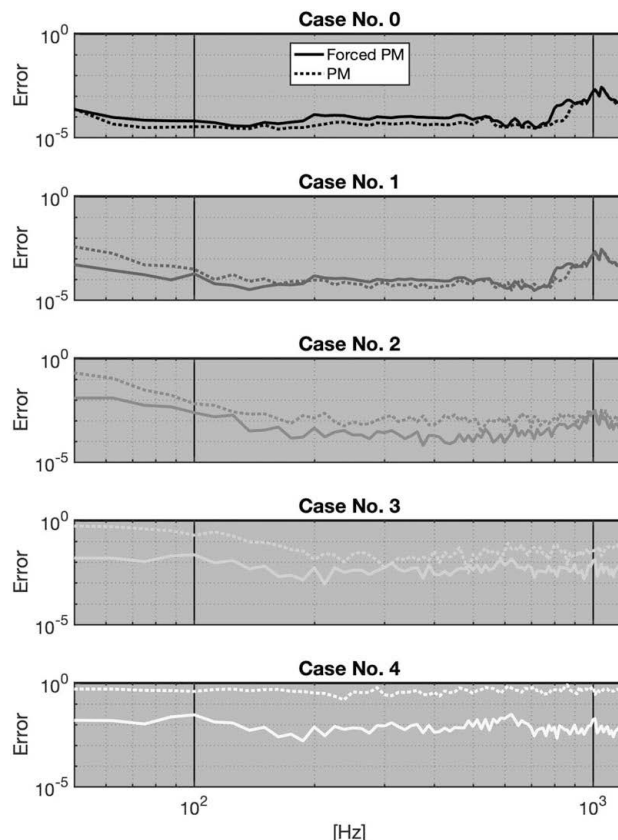


Fig. 6. Resulting E_{rr} for the FPM (bold) and the PM (dashed) using measured transfer functions for five different levels of Gaussian noise introduced in the TF.

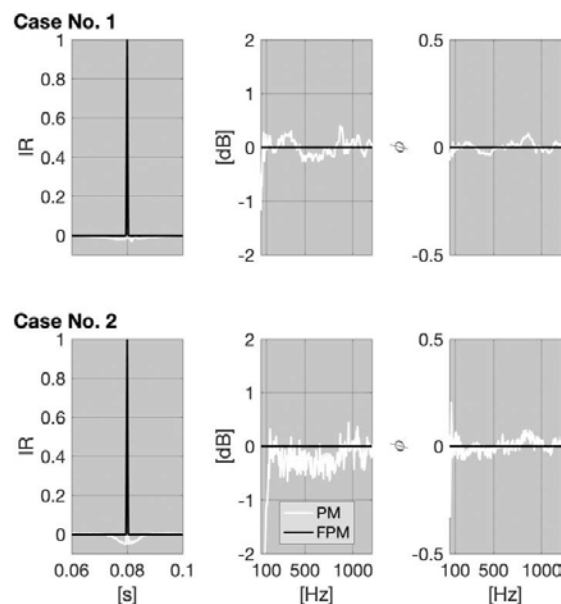


Fig. 7. Spk. ③ resulting filter using FPM (black) and PM (white) under the influence for two Noise cases. First column time IR shifted to the center of the IR and second column corresponds to the modulus of the frequency response and the third column to the phase of the frequency response.

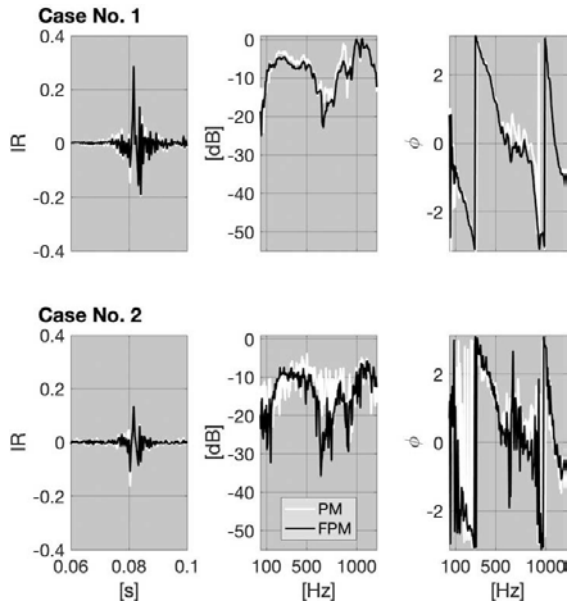


Fig. 8. Spk. ④ resulting filter using FPM (black) and PM (white) under the influence for two Noise cases. First column time IR shifted to the center of the IR and second column corresponds to the modulus of the frequency response and the third column to the phase of the frequency response.

one proceeded to evaluate the effect of noise in other filters rather than the “forced ones.”

For example, in Fig. 8, it is possible to observe the effect of noise in the filter associated to Spk. ④. When comparing the resulting filters for Spk. ④ without extra noise (Fig. 4) and the resulting filters with low levels of noise are present (Case No. 1 in Fig. 8), it is possible to observe that both methods relatively maintain their differences as observed in that there are some differences below 100 Hz and a steeper attenuation for the PM at 906 Hz. However, for the SNR Case No. 2 the filter calculated by the PM became very noisy with a very complicated phase response below 300 Hz where the SNR is very small. On the other hand, in spite of also changing its shape, the FPM presents a smoother response than the PM method both in amplitude and phase.

5 CONCLUSIONS AND FUTURE WORK

The FPM was introduced as an alternative of the PM methods when specific filter responses are forced into the calculation. This study showed its analytical demonstration and an application in a typical TI case where two seats are placed next to each other.

When comparing the performance of the PM and FPM, both result in the same AC, the same E_{ff} , and very similar E_{rr} . On the other hand, the FPM presented a better conditioning and it took $\approx 24\%$ less time to calculate the filters. The FPM not only allows us to have a specific response in some filters but it can also be used to reduce the number of variables involved in the calculation, which leads to a faster calculation and smaller sensitivity to noise.

On the one hand, the FPM can be thought of as the PM that has been already fed with a part of the solution; there is a part of the problem that does not need to be solved. For the headrest PSZ application, this results in a new—also twofold—problem: in the DZ it is necessary to cancel what is coming from the forced loudspeakers (hence the “negative” sign in the p_{Tc} in Eq. (22)) and it needs to do this without disturbing the BZ (the potential energy minimization term in Eq. (23)). The main drawback of the FPM is that it is necessary to have some knowledge of the solution, i.e., the “good solution” needs to be fed to the algorithm in order to profit all the advantages of the FPM reviewed in this paper. Otherwise, the forcing can be seen as a sort-of imposition to the algorithm and it will severely affect its performance.

On the other hand, the FPM can also be seen as a more constrained PM that results in a reduction/simplification of the “degrees of freedom” that the algorithm has to solve the problem (the size of $\mathbf{q}$ is by definition smaller than the size of $\mathbf{q}_c$). Evidence of this can be seen in Fig. 4 from the differences between the calculated filters. Moreover the effect of this simplification can be the explanation of why the FPM is better conditioned than the PM (Fig. 2) and also why (in Fig. 6) without noise the E_{rr} is slightly worse but also why with noise the reproduction is better even under unrealistic levels of noise (e.g., Case No. 3 or 4). In addition, it can also explain why both formulations under the presence of noise result in such different filters (see Fig. 8 Case No. 2) in spite of being “corrupted” by identical noises.

The FPM was explained in this paper using a Frequency domain formulation, where the complex weights are calculated for particular frequencies; hence the calculation needs to be repeated for the bandwidth one wants to control. As was briefly mentioned, the FPM principle—subtracting the contribution of the forced filters from the target pressure and reducing the TF matrix—can also be applied to other formulations as long as the least-squared error is being minimized. Take for example the time domain formulation used in Simón Gálvez et al. in 2015 [5].

There are several reasons why the FPM is particularly interesting for the TI. In an “end of the line situation,” the time available for the calibration of the system is short and the conditions are not always ideal for a TF measurement with large SNR. The FPM presents smaller sensitivity to noise than the PM; it ensures the shape of the filters associated to the reproduction (the closest to the BZ) and reduces the number of filters to be calculated and stored. For example, if one decides to anticipate “all the possible changes” and store all the filter combinations, the FPM allows the system to store fewer filters than with the PM (25% for the studied application), reducing the cost of storage. Moreover, once the product is out of the factory, due to the acoustic variability of the TI environments, it is more than realistic to assume that some adaptation or re-calibration will be needed to be performed “on the go.” Therefore the earnings on robustness as well as the reduction of computational capabilities needed to be installed to offer a considerable advantage. For example, if an adaptive system that re-calculates the filters

is used, the FPM reduces the time taken to obtain the time taken by the system to obtain the new filter combination (in this paper $\approx 24\%$ faster).

5.1 Future Work

Firstly it is necessary to take these results one step forward toward the real TI application; i.e., the same study must be carried out under non-anechoic conditions to evaluate if the FPM is still performing as well as the classical PM in more acoustically complex environments. In addition a δ function is desired in the reproduction sources in order to avoid any temporal artifacts [26]. Nevertheless a series of psychoacoustic tests could be performed to evaluate if the improvements on the reproduction are noticeable. For example, using the two combinations of filters studied in this paper: those obtained under laboratory conditions (Fig. 4), and those with noisy TFs (Fig. 7).

As mentioned in the introduction, the higher part of the hearing spectrum is assumed to be handled by a different technology. Therefore it might also be interesting to consider inside the optimization the needed crossover to merge these two different solutions.

Furthermore, a more thorough analysis could be performed to fully evaluate the effect of the loudspeaker-reduction on the computational cost. For example taking as a reference the work presented in [19], one could add to the calculation time the working memory requirements and evaluate them using different array-sizes, different filter lengths, different formulations, etc., but this is out of the scope of this paper.

As was previously mentioned, it is not only possible to apply the FPM in the case where a piece of the answer is known beforehand (as in this study) but also to use it to add the filter responses as a new constraint. Therefore it can be interesting to evaluate its advantages using different systems with different goals such as controlling the directivity of an array [27–29].

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